



Paper ID : 260918

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Subject Code: BMC055

Roll No:

MCA

(SEM. IV) THEORY EXAMINATION 2025-26

QUANTUM COMPUTING

TIME: 3 HRS

M.MARKS: 70

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt all questions in brief.**

2x7 = 14

|     |                                                                            |     |    |
|-----|----------------------------------------------------------------------------|-----|----|
| (a) | Define a qubit and state how it differs from a classical bit.              | CO1 | K2 |
| (b) | State any two postulates of quantum mechanics used in quantum computation. | CO1 | K1 |
| (c) | What is the role of measurement in a quantum circuit?                      | CO2 | K2 |
| (d) | Define universal quantum gates with any two examples.                      | CO2 | K2 |
| (e) | List the essential conditions required for building a quantum computer.    | CO2 | K2 |
| (f) | Define quantum noise and give one suitable example.                        | CO4 | K1 |
| (g) | What is Shannon entropy? Mention its significance in information theory.   | CO5 | K2 |

## SECTION B

**2. Attempt any three of the following:**

7 x 3 = 21

|     |                                                                                                                                                           |     |    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Explain the fundamental concepts of quantum computation with reference to qubits, superposition, quantum information and postulates of quantum mechanics. | CO1 | K2 |
| (b) | Describe quantum circuits. Explain single-qubit operations, controlled operations and measurement with suitable examples.                                 | CO2 | K3 |
| (c) | Explain the Quantum Fourier Transform. Discuss its importance in phase estimation and quantum algorithms.                                                 | CO3 | K3 |
| (d) | Explain quantum operations and quantum noise. Discuss how the formalism represents transformations and noise in quantum systems.                          | CO4 | K3 |
| (e) | What is Von Neumann entropy? State its importance in quantum information.                                                                                 | CO5 | K2 |

## SECTION C

**3. Attempt any one part of the following:**

7 x 1 = 7

|     |                                                                                                                                                       |     |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Distinguish between classical computation and quantum computation. Explain how quantum computation can make certain computational problems tractable. | CO1 | K2 |
| (b) | Explain quantum algorithms and discuss how they differ from classical algorithms with reference to computational complexity.                          | CO1 | K2 |

**4. Attempt any one part of the following:**

7 x 1 = 7

|     |                                                                                                                            |     |    |
|-----|----------------------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Explain Grover's quantum search algorithm for an unstructured database. How does it provide speedup over classical search? | CO2 | K3 |
| (b) | Write a detailed note on quantum counting and its relationship with quantum search algorithms.                             | CO2 | K3 |

**5. Attempt any one part of the following:**

7 x 1 = 7

|     |                                                                                                                                                                                                                                         |     |    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Explain how a team can develop a medium-scale simulation program for an optical photon quantum computer using suitable collaborative development tools such as Git or GitHub.                                                           | CO3 | K3 |
| (b) | Discuss the steps involved in developing a simulation program for ion-trap or nuclear magnetic resonance-based quantum computers as part of a cooperative team project. Also mention the role of version control in project management. | CO3 | K3 |

**6. Attempt any one part of the following:**

7 x 1 = 7

|     |                                                                                                               |     |    |
|-----|---------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Explain classical noise and Markov processes. How are they related to quantum noise and quantum operations?   | CO4 | K3 |
| (b) | Discuss distance measures for quantum information and explain their significance in comparing quantum states. | CO4 | K3 |

**7. Attempt any one part of the following:**

7 x 1 = 7

|     |                                                                                                                                                       |     |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| (a) | Explain the theory of quantum error correction. Discuss Shor code and its significance in protecting quantum information.                             | CO5 | K6 |
| (b) | Write a detailed note on stabilizer codes and fault-tolerant quantum computation. Explain the role of entropy and entanglement as physical resources. | CO5 | K6 |